Structure-Exploiting Factor-Risk Quadratic Programming for One-Shot and Ladder Portfolio Construction

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Abstract. Equity long–short portfolio construction repeatedly solves convex quadratic programs whose covariance decomposes into a low-rank factor component and a diagonal specific component, $\boldsymbol{\Sigma} = \mathbf{B}\mathbf{F}\mathbf{B}^\top + \operatorname{diag}(\mathbf{d})$. Modelled through general-purpose expression graphs and solved as generic conic programs, the problem spends much of its runtime assembling and canonicalising. We describe a primal–dual Mehrotra predictor–corrector interior-point method that keeps factor risk factored through the Newton step and reuses static constraint data across rebalance ladders. Slack elimination condenses the Karush–Kuhn–Tucker system into a diagonal-plus-low-rank matrix of rank $r = k + m$, with k risk factors and m two-sided exposure bands. The Woodbury identity and a small Schur complement reduce per-iteration linear algebra from $O(n^3)$ to $O(nr^2 + r^3)$. On a synthetic 1,000-security book with $r = 43$, single-thread solve time falls from 1,275.5 ms under CVXPY with Clarabel and 97.7 ms under a hand-built sparse Clarabel model to 25.3 ms, with an objective gap of 5.3×10^{-6}. Across a 36-rung ladder at $n = 500$ the method is $18.9\times$ faster than rebuilding through CVXPY and $3.1\times$ faster than the sparse model. In a synthetic 12-month run of a four-stage market-neutral construction sequence, end-to-end runtime falls from 8.73 s to 2.93 s. The speedup comes with small but nonzero differences in weights, which we report.

Problem Formulation

Cross-sectional equity strategies trade off expected return against factor and specific variance, subject to limits on positions, exposures, leverage, and turnover. A research process then solves the same problem many times: across gross targets, support sizes, and formation dates. The numerical kernel is a convex QP in weights $\mathbf{x} \in \mathbb{R}^n$:

$$\begin{aligned} \min_{\mathbf{x}} \quad & \frac{1}{2}\mathbf{x}^\top \mathbf{P}\mathbf{x} + \mathbf{q}^\top \mathbf{x} + c_0 \\ \text{s.t.} \quad & \boldsymbol{\ell} \le \mathbf{x} \le \mathbf{u}, \\ & \mathbf{g}_\ell \le \mathbf{G}\mathbf{x} \le \mathbf{g}_u, \quad \mathbf{A}\mathbf{x} = \mathbf{b}. \end{aligned} \tag{1}$$

Box bounds encode position caps, liquidity limits, and fixed signs. $\mathbf{G}$ holds exposure bands, and $\mathbf{A}$ holds equalities such as gross leverage for a fixed sign vector. For exposures $\mathbf{B} \in \mathbb{R}^{n\times k}$, factor covariance $\mathbf{F}$, and specific variances $\mathbf{d}$, the covariance is

$$\boldsymbol{\Sigma} = \mathbf{B}\mathbf{F}\mathbf{B}^\top + \operatorname{diag}(\mathbf{d}). \tag{2}$$

With risk aversion γ and quadratic trading-cost weights $\boldsymbol{\tau}_2$, the operator keeps a factored form,

$$\mathbf{P} = \mathbf{U}\operatorname{diag}(\mathbf{c})\,\mathbf{U}^\top + \operatorname{diag}(\mathbf{p}_d), \tag{3}$$

where $\mathbf{F} = \mathbf{V}\boldsymbol{\Lambda}\mathbf{V}^\top$, $\mathbf{U} = \mathbf{B}\mathbf{V}\boldsymbol{\Lambda}^{1/2}$, $\mathbf{c} = \gamma \mathbf{1}_k$, and $\mathbf{p}_d = \gamma\mathbf{d} + \boldsymbol{\tau}_2$.

Through CVXPY (Diamond and Boyd, 2016) with Clarabel (Goulart and Chen, 2024), two overheads recur on every solve: forming dense $\boldsymbol{\Sigma}$ costs $O(n^2 k)$, and canonicalising the dense quadratic form leads to an $O(n)$-dimensional factorisation per iteration. A hand-built sparse model with auxiliary factor variables $\mathbf{f} = \mathbf{U}^\top\mathbf{x}$ avoids most of this, and it is the fairer baseline, but it repeats cone setup and general-purpose factorisation on each call.

Table 1. Single-thread benchmarks on a 2-vCPU Linux VM (tol $= 10^{-8}$, medians of repeated runs, BLAS pinned to one thread, one core). Problems are synthetic. The baseline is CVXPY with Clarabel or, for the 100-name book, SLSQP, and accuracy is measured against it; "sparse" is the hand-built sparse Clarabel model; dashes mean not recorded.

Workload	Solver path	Median (ms)	vs. baseline	vs. sparse	Objective gap	Max $\lvert\Delta x_i\rvert$
One-shot ($n = 300$, $r = 43$)	CVXPY + Clarabel	88.3	1.0×	—	0	0
	Sparse Clarabel	26.4	3.3×	1.0×	2.1e-9	1.2e-5
	This method (12 iterations)	**10.0**	**8.8**×	**2.6**×	2.1e-6	4.4e-4
One-shot ($n = 1000$, $r = 43$)	CVXPY + Clarabel	1,275.5	1.0×	—	0	0
	Sparse Clarabel	97.7	13.1×	1.0×	2.1e-10	2.3e-7
	This method (18 iterations)	**25.3**	**50.4**×	**3.9**×	5.3e-6	5.3e-4
Ladder ($n = 500$, 36 solves, per solve)	CVXPY rebuild	285.5	1.0×	—	0	0
	Sparse Clarabel	47.2	6.0×	1.0×	—	—
	This method, compiled	**15.1**	**18.9**×	**3.1**×	3.2e-6	5.5e-4
Market-neutral book ($n = 200$)	CVXPY + Clarabel	46.3	1.0×	—	0	0
	This method	**9.8**	**4.7**×	—	5.3e-7	4.5e-4
Signed book ($n = 100$, dense risk)	SciPy SLSQP	150.6	1.0×	—	0	0
	This method	**38.6**	**3.9**×	—	—	7.4e-5

Factored Interior-Point Kernel

Rank-one augmentations. Penalties on linear aggregates preserve (3). A dollar-neutrality penalty $\frac{1}{2}\rho_{\text{net}}(\mathbf{1}^\top\mathbf{x})^2$ appends the column $\mathbf{1}$ to $\mathbf{U}$ and the weight ρ_{net} to $\mathbf{c}$. A target penalty $\frac{1}{2}\lambda_j(\mathbf{v}_j^\top\mathbf{x} - \beta_{0,j})^2$ appends $\mathbf{v}_j$ and λ_j, shifts $\mathbf{q}$ by $-\lambda_j\beta_{0,j}\mathbf{v}_j$, and adds $\frac{1}{2}\lambda_j\beta_{0,j}^2$ to c_0.

Primal–dual system. We introduce positive slacks $\mathbf{s}_{\text{hi}}, \mathbf{s}_{\text{lo}}, \mathbf{s}_{gu}, \mathbf{s}_{gl}$, multipliers $\mathbf{z}_{\text{hi}}, \mathbf{z}_{\text{lo}}, \mathbf{z}_{gu}, \mathbf{z}_{gl}$, and equality multipliers $\mathbf{y}$. Infinite bounds are masked out. The residuals are

$$\begin{aligned} \mathbf{r}_d &= \mathbf{P}\mathbf{x} + \mathbf{q} + (\mathbf{z}_{\text{hi}} - \mathbf{z}_{\text{lo}}) + \mathbf{G}^\top(\mathbf{z}_{gu} - \mathbf{z}_{gl}) + \mathbf{A}^\top\mathbf{y}, \\ \mathbf{r}_{p,\text{hi}} &= \mathbf{x} + \mathbf{s}_{\text{hi}} - \mathbf{u}, \qquad \mathbf{r}_{p,\text{lo}} = -\mathbf{x} + \mathbf{s}_{\text{lo}} + \boldsymbol{\ell}, \\ \mathbf{r}_{p,gu} &= \mathbf{G}\mathbf{x} + \mathbf{s}_{gu} - \mathbf{g}_u, \quad \mathbf{r}_{p,gl} = -\mathbf{G}\mathbf{x} + \mathbf{s}_{gl} + \mathbf{g}_\ell, \\ \mathbf{r}_e &= \mathbf{A}\mathbf{x} - \mathbf{b}, \end{aligned} \tag{4}$$

where $\mathbf{P}\mathbf{x} = \mathbf{U}(\mathbf{c} \odot (\mathbf{U}^\top\mathbf{x})) + \mathbf{p}_d \odot \mathbf{x}$ costs $O(nk)$.

Woodbury normal equations. With barrier weights $\mathbf{w}_\bullet = \mathbf{z}_\bullet \oslash \mathbf{s}_\bullet$, pivoting out $(\Delta\mathbf{s}, \Delta\mathbf{z})$ reduces each Newton step to

$$\mathbf{M}\,\Delta\mathbf{x} + \mathbf{A}^\top\Delta\mathbf{y} = \mathbf{r}_{\text{rhs}}, \qquad \mathbf{A}\,\Delta\mathbf{x} = -\mathbf{r}_e, \tag{5}$$

where

$$\begin{aligned} \mathbf{M} &= \mathbf{P} + \operatorname{diag}(\mathbf{w}_{\text{hi}} + \mathbf{w}_{\text{lo}}) + \mathbf{G}^\top\operatorname{diag}(\mathbf{w}_{gu} + \mathbf{w}_{gl})\mathbf{G} \\ &= \mathbf{H} + \mathbf{U}_{\text{aug}}\operatorname{diag}(\mathbf{c}_{\text{aug}})\,\mathbf{U}_{\text{aug}}^\top. \end{aligned} \tag{6}$$

Here $\mathbf{H} = \operatorname{diag}(\mathbf{p}_d + \mathbf{w}_{\text{hi}} + \mathbf{w}_{\text{lo}} + \varepsilon\mathbf{1})$ with $\varepsilon = 10^{-12}$, the matrix $\mathbf{U}_{\text{aug}} = [\mathbf{U}, \mathbf{G}^\top]$ is fixed before the iteration loop, and $\mathbf{c}_{\text{aug}} = \max([\mathbf{c};\, \mathbf{w}_{gu} + \mathbf{w}_{gl}], \varepsilon)$. By the Sherman–Morrison–Woodbury identity (Woodbury, 1950), applying $\mathbf{M}^{-1}$ needs only an $r \times r$ capacitance system:

$$\mathbf{K} = \operatorname{diag}(\mathbf{c}_{\text{aug}})^{-1} + \mathbf{U}_{\text{aug}}^\top\mathbf{H}^{-1}\mathbf{U}_{\text{aug}}, \tag{7}$$

$$\mathbf{M}^{-1}\mathbf{V} = \mathbf{H}^{-1}\mathbf{V} - \mathbf{H}^{-1}\mathbf{U}_{\text{aug}}\mathbf{K}^{-1}\mathbf{U}_{\text{aug}}^\top\mathbf{H}^{-1}\mathbf{V}. \tag{8}$$

Assembling $\mathbf{K}$ costs $O(nr^2)$ and factorising it costs $O(r^3)$. With p equalities, the $p \times p$ Schur complement $\mathbf{S} = \mathbf{A}\mathbf{M}^{-1}\mathbf{A}^\top$ gives

$$\begin{aligned} \Delta\mathbf{y} &= \mathbf{S}^{-1}\left(\mathbf{A}\mathbf{M}^{-1}\mathbf{r}_{\text{rhs}} + \mathbf{r}_e\right), \\ \Delta\mathbf{x} &= \mathbf{M}^{-1}\mathbf{r}_{\text{rhs}} - (\mathbf{M}^{-1}\mathbf{A}^\top)\Delta\mathbf{y}. \end{aligned} \tag{9}$$

Predictor–corrector step. Following Mehrotra (1992), each iteration computes an affine-scaling direction, the maximum step that keeps slacks and multipliers at no less than 1% of their current values, and the centering parameter $\sigma = \min(\max(\mu_{\text{aff}}/\mu, 0), 1)^3$. The combined step replaces the complementarity term with $\mathbf{s} \odot \mathbf{z} + \Delta\mathbf{s}_{\text{aff}} \odot \Delta\mathbf{z}_{\text{aff}} - \sigma\mu\mathbf{1}$ and reuses $\mathbf{K}$, $\mathbf{M}^{-1}\mathbf{A}^\top$, and $\mathbf{S}$. Convergence is declared when all residual norms and μ are below tol, by default 10^{-8}.

Outer Loops and Ladder Compilation

Ladder compilation. Across a ladder, $\mathbf{U}$, $\mathbf{G}$, $\mathbf{A}$, $\mathbf{U}_{\text{aug}}$, and the active-bound masks are invariant, while $\mathbf{q}$ and $\mathbf{b}$ vary. We cache the static data once, and each later rung enters the interior-point loop directly.

Ex-ante risk cap. For a hard budget $\mathbf{x}^\top\boldsymbol{\Sigma}\mathbf{x} \le \sigma_{\max}^2$, we solve at the baseline γ. If the budget is exceeded, we bracket the multiplier shift λ by doubling and bisect on $\gamma + \lambda$. Each inner solve warm-starts the primal vector and the equality dual, and resets barrier slacks and inequality duals to avoid numerical blow-up on constraints that were nearly active.

L1 turnover. A penalty $\sum_i \tau_{1,i}|x_i - x_{i,\text{prev}}|$ is handled without $2n$ splitting variables by iteratively reweighted least squares. With previous trades $\mathbf{q}^{(t-1)}$ the smoothed majoriser weights are

$$a_i^{(t)} = \frac{\tau_{1,i}}{\sqrt{\left(q_i^{(t-1)}\right)^2 + \epsilon^2}}, \qquad \epsilon = 10^{-8}, \tag{10}$$

which are added to the diagonal. Each pass remains a diagonal-plus-low-rank QP of unchanged dimension and rank.

Soft bands. One-sided soft bands default to a quadratic slack penalty with $\rho_{\text{soft}} = 10^4$; exact linear slacks are available at the cost of more iterations on degenerate books.

Fast Screening Estimators

Loose-tolerance preview. Stopping at tol $= 10^{-3}$ and six iterations gives an approximate solution for screening and warm starts (Table 2, from a separate timing run). The objective gap is 2.8–3.3×10^{-3} and the maximum weight difference is at most 1.9×10^{-2}, so a preview ranks candidates but is not a book. Allowing up to 16 iterations tightens these to 7.5×10^{-5} and 4×10^{-3}.

Table 2. Exact solve (tol $= 10^{-8}$) vs. six-iteration preview (tol $= 10^{-3}$), in milliseconds.

n	Exact	Preview	Speedup	Obj. gap	Max $\lvert\Delta x\rvert$
300	10.0	3.7	2.7×	3.3×10^{-3}	1.6×10^{-2}
1000	24.7	4.9	5.0×	2.8×10^{-3}	1.9×10^{-2}

Constraint-contact screen. Motivated by jamming in constrained perceptrons (Franz and Parisi, 2016), we define the contact ratio of a sign-fixed candidate book:

$$\rho_{\text{contact}} = \frac{\#\{i : x_i \text{ at a bound}\} + \#\{\text{active bands}\}}{n_{\text{selected}}}. \tag{11}$$

On 16 labelled formations from our market-neutral research, all successful first rungs of a gross ladder had $\rho_{\text{contact}} < 0.47$, while the two failures had 0.59 and 0.65. A two-parameter logistic with threshold 0.53 and slope 100 turns this into a ranking score. The two failures come from a single calendar month, so this is a screening heuristic and not a probability.

Empirical Scaling

Table 1 separates two effects at $n = 1{,}000$. Removing the expression graph and dense canonicalisation gives $13.1\times$ (1,275.5 ms to 97.7 ms). Replacing general conic factorisation with the 43×43 Woodbury system gives a further $3.9\times$ (to 25.3 ms), for $50.4\times$ in total. With rank fixed, larger signed stage-0 books with no minimum position solve in about 80 ms at $n = 2{,}000$ and about 200 ms at $n = 5{,}000$.

In a synthetic 12-month run of a four-stage market-neutral construction sequence ($n = 400$, 10 sectors, 5 styles; Table 3), a single base QP falls from 86.8 ms to 20.1 ms ($4.3\times$, maximum weight difference 3.5×10^{-7}). Supported problems use the solver, and unsupported features fall back to Clarabel. The last two rows of Table 3 also substitute approximations: a fixed-sign preview in the first pass and symmetric L1 trade costs in the last stage. They trade exactness for speed, with maximum weight differences near 3×10^{-3} and 2×10^{-3}.

Table 3. Four-stage construction runtime in seconds, 12 synthetic formations ($n = 400$).

Solver path	Stage 1	Stage 2	Stage 3	Stage 4	Total
Clarabel baseline	0.690	2.572	2.652	2.812	8.726
This method, final QPs	0.160	1.714	1.779	2.849	6.502
+ fixed-sign preview	0.157	0.706	0.694	2.883	4.439
+ symmetric L1 costs	**0.160**	**0.673**	**0.698**	**1.399**	**2.930**

Limitations

First, all benchmarks use synthetic problems on one machine, so absolute times will differ on real data and other hardware. Second, this solver is not a basis for any published performance figure: QUARCC Purple's submitted book was solved with Clarabel, and the four-stage result above measures runtime rather than investment outcomes. Third, the method assumes sign-fixed weights, so free-sign support selection with minimum positions still needs an outer ladder or a first-stage sign assignment. Fourth, L1 turnover reaches the exact objective to roughly 10^{-5}, and the default quadratic soft-band penalty explains weight differences near 5×10^{-4}. Fifth, the contact screen rests on two failures from one month.

Availability

The solver is an internal QUARCC research tool and is not publicly released at this time.

References

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